

Fainting

By **Andrea D. Thompson**, MD, PhD, University of Michigan;
Michael J. Shea, MD, Michigan Medicine at the University of Michigan
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Light-headedness (near syncope) is a sense that one is about to faint.

Fainting (syncope) is a sudden, brief loss of consciousness during which the person falls to the ground or slumps in a chair followed by a return to consciousness. The person is motionless and limp and may have cool legs and arms, a weak pulse, and shallow breathing.

Some people feel light-headed or dizzy before they faint. Others may have nausea, sweating, blurred vision or tunnel vision, tingling of lips or fingertips, chest pain, or palpitations. Less often, people faint suddenly, without any warning symptoms.

Seizures, which are a disturbance of the brain's electrical activity, and cardiac arrest, in which the heart completely stops beating, can cause loss of consciousness but are not considered fainting. However, in some people who faint, muscles briefly jerk involuntarily, resembling a seizure.

Fainting can occur in people of any age, but dangerous causes of fainting are more common among older adults.

Causes of Fainting

A person cannot lose consciousness unless brain function is generally disturbed. This disturbance usually occurs because overall blood flow to the brain is reduced. Sometimes, however, blood flow is adequate but the blood does not contain sufficient oxygen or glucose (blood sugar), which the brain needs to function.

Blood flow to the brain can be reduced in several ways. Most often, the cause is something that interferes with the normal return of blood to the heart (and thus reduces blood flow out from the heart). Less often, the cause is a disorder that

interferes with blood pumping (typically a heart disorder). Although strokes reduce blood flow to the brain, they only reduce flow to part of the brain. Thus, strokes rarely cause fainting except for the few strokes that involve the part of the brain that maintains consciousness.

The **most common causes** of fainting are

- Strong emotion (such as fear, pain, or sight of blood)
- Coughing or straining to pass stool or urine
- Prolonged standing
- Standing up suddenly
- Pregnancy
- Use of certain medications
- Idiopathic (meaning that the cause cannot be determined)

These common causes nearly always cause fainting only when people are standing up. When they fall down, blood flow to the brain is increased, quickly restoring consciousness, although people may not feel completely normal for a few minutes to a few hours. Some people feel tired or exhausted for several hours. These causes tend not to be serious unless people are injured when they fall.

Most of these causes involve decreased return of blood to the heart. Strong emotion (particularly that triggered by the sight of blood) or pain can activate the vagus nerve. Vagus nerve activation widens blood vessels, reducing the return of blood to the heart, and slows the heart rate. Both of these factors cause light-headedness and sometimes fainting (called vasovagal syncope or neurocardiogenic syncope).

Straining during bowel movements or urination or coughing increases chest pressure. Increased chest pressure can activate the vagus nerve and also reduce the return of blood to the heart—2 factors that may cause fainting.

Healthy people may faint when standing still for a long time (most common in soldiers, a phenomenon called parade ground syncope), because the leg muscles have to be active to help return blood to the heart.

Sitting or standing up too quickly can cause fainting, because the change in position causes blood to pool in the legs, resulting in a fall in blood pressure. Normally, the body quickly increases the heart rate and constricts blood vessels to maintain blood pressure. If the body does not compensate in these ways, light-headedness is common and fainting may rarely occur. Certain brain and spinal cord disorders, prolonged bed rest, and certain medications (particularly those used to treat high

blood pressure) can interfere with this compensation and lead to [fainting when standing up](#).

Hormonal changes in early pregnancy sometimes lead to fainting.

Low blood sugar ([hypoglycemia](#)) initially causes confusion, light-headedness, shakiness, and other symptoms, but if hypoglycemia is severe or prolonged, people can lose consciousness. Because these other symptoms usually occur before fainting, people with hypoglycemia usually have some warning before they faint. Usually, the cause of hypoglycemia is use of medications for diabetes, particularly insulin. Rarely, people have a tumor that secretes insulin.

Less common but more serious causes include

- [Heart valve disorders](#) (most commonly, involving the aortic valve)
- A heart rate that is too fast or too slow
- Blockage of an artery to the lungs by a blood clot ([pulmonary embolism](#))
- [Heart attack](#) or other heart muscle disorders

Heart valve disorders can block blood from leaving the heart. A very fast heart rate may not allow the heart enough time to refill with blood, so less blood is pumped. A very slow heart rate may not pump enough blood. Blood clots in the lungs can keep the heart from pumping enough blood. People with a heart attack rarely have fainting when the heart attack occurs (more common in older people). Other uncommon heart muscle disorders called [cardiomyopathies](#) can cause fainting, particularly during exercise, typically because of an [abnormal heart rhythm](#).

Although most strokes do not cause fainting, a [stroke](#) or [transient ischemic attack](#) (TIA) that involves certain blood vessels at the base of the brain (posterior circulation stroke) can cause fainting. Similarly, a [migraine](#) that involves these blood vessels sometimes causes fainting.

Evaluation of Fainting

If possible, someone who witnessed the faint should provide the doctor with a description of the event because the person who fainted may not remember.

Warning signs

In people who have fainted, certain symptoms and characteristics are cause for concern. They include

- Fainting during exercise
- Several episodes within a short time
- Sudden fainting without any warning symptoms or any apparent trigger
- Fainting preceded or followed by possible heart symptoms such as chest pain, palpitations, or shortness of breath
- Older age
- Significant injury as a result of fainting
- Family history of sudden unexpected death, fainting during exercise, or repeated episodes of fainting or seizures for which a cause has not been found

When to see a doctor

Although most causes of fainting are not serious, a doctor's evaluation is needed to distinguish serious causes from relatively harmless ones. People who fainted should see a doctor right away, especially if they have any warning signs.

What the doctor does

Doctors first ask questions about the person's symptoms and medical history. Doctors then do a physical examination. What doctors find during the history and physical examination often suggests a cause of the fainting and the tests that may need to be done.

Doctors ask about the events leading up to the fainting episode. They ask whether the person was exercising, arguing, or in a potentially emotional situation. They ask whether the person was lying or standing, and, if standing, for how long. They also ask about symptoms that occurred immediately before or after the event, including whether the person felt light-headed or dizzy or experienced nausea, sweating, blurred or tunnel vision, tingling of lips or fingertips, chest pain, or palpitations.

Doctors also ask any witnesses to describe the episode. A sudden, abrupt faint without any warning symptoms or apparent trigger suggests a heart disorder. If fainting is preceded by a brief period of symptoms such as light-headedness, nausea, yawning, blurred vision, or sweating and occurs during a painful or unpleasant situation, it is more likely vasovagal syncope, which is not dangerous.

Doctors ask about previous episodes of dizziness or fainting and about other disorders, medications the person is taking, or symptoms that may be related to fainting. Doctors also check the person for injuries resulting from the fainting episode.

Then doctors measure the person's vital signs. Heart rate and blood pressure are measured with the person lying down and after 1 to 3 minutes of standing. Doctors listen to the heart for signs of an abnormal heart valve or abnormal blood flow. They do a neurologic examination for signs of a stroke.



Some Causes and Features of Fainting

Cause	Common Features*	Tests†
Serious causes		
Heart valve disorders, such as aortic stenosis , mitral stenosis , or failure of an artificial heart valve	Fainting during or after exercise, followed by a prompt recovery	Echocardiography (ultrasound study of the heart)
Cardiomyopathy (disorders affecting heart muscle), particularly hypertrophic cardiomyopathy	In young or old people Often in people who are known to have had a heart murmur	
	Fainting without warning, followed by recovery immediately after awakening	
A very slow heart rate (typically less than 35 beats per minute), more common in older adults	Fainting that may occur in any position Sometimes in people taking certain medications, especially medications used to treat heart disorders such as abnormal heart rhythms	ECG, sometimes continuous ambulatory ECG (using a Holter monitor or longer recording with an event monitor)
A very rapid heart rate (typically over 150 beats per minute)	(antiarrhythmic medications) or angina (such as beta-blockers and/or calcium channel blockers)	Sometimes blood tests to measure electrolytes such as sodium and potassium
Pulmonary embolism (blockage of an artery)	Often sharp pain when breathing in,	



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